"On my honor, I have neither given nor received any unathorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues"

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1) $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \bigcirc\!\!\!\!\beta & 3 & \alpha^2 \\ 1 & \boxed{14} & -6 & 2 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \boxed{-1} & 1 & 0 \\ 0 & \boxed{k} & 0 & 1 \end{bmatrix}$

$\alpha^2 = 0$

$\boxed{\alpha = 0}$

$\boxed{\gamma = 0}$

$k = 14$

$\boxed{\beta = -1}$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$

$\boxed{m = 2}$ geliyor

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 1 \end{bmatrix}$

↓

buna burada şuan bakmadık

a) find the unknown entries - - - -

$\alpha = 0$, $\gamma = 0$, $k = 12$, $m = 2$

b) Find the eliminated upper triangle matrix U ?

upper triangle demek $\begin{bmatrix} 0 & \diagdown \\ 0 & 0 \end{bmatrix}$ → köşegen çek, alt tarafı 0

$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & 5 & -9\gamma & 1 \\ 0 & 0 & 3 & \alpha^2 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ → upper triangle matrix

2) A geometry in 'k' dimensional space is given by $x=5y$ expression, where 'x' and 'y' represent the axis. The intersection with another geometry gives a volume.

b) kürenin içinde bir plane
→ kesit

Yol

c) $x+4y-z=6$

$$\begin{bmatrix} 1 & 4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 6$$

$\downarrow$ pivot $\downarrow$ free $\downarrow$ free

intersection için inceleyelim

$$\begin{cases} x+4y-z=6 \\ x-5y=0 \end{cases}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \alpha \begin{pmatrix} 5 \\ 1 \\ 9 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix}$$

Önce yaptıklarımı sildim, buraya sığmadı

Umarım sorun olmadım

Her vektör space zero vektörünü içerir ama her zero vektörünü içeren vector space değildir!

$x+4y-z=6$

Bu bir vector space değildir çünkü zero vektörünü içermiyor

2D inside 3D

3) Suppose u is a unit in $\mathbb{R}^n$. This problem is about the n by n symmetric matrix $H = I - 2uu^T$

If u is a unit vector. Lets say $v = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \rightarrow \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = 1$

$H = I - 2uu^T$

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \cdot \underset{2\times 1}{\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}} \cdot \underset{1\times 2}{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}}$$

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$H = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

a) Show directly $H^2 = I$ (H^2'yi bulmak için H'ın içindekilerin karesini almayız, yanlış çıkar. İki kere çarpmak gerek)

$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$$

→ eşitlik sağlandı :)

b) Show that H is an orthonormal matrix

Genel kural

1. yöntem → $A^{-1} = A^T$ ise ortogonal olur.

2. yöntem → $A^T \cdot A = I$

$\boxed{H^2 = I}$

$H \cdot H^T = H^T \cdot H = I$ → orthonormal matrix

2. yöntemi uygularsak →

$$H^T \cdot H \overset{?}{=} I \Rightarrow \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$$

3'ün devamı

c) ① One eigenvect of H is u itself. Find corresponding eigenvalue.

$H = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

$|H - hI| = 0$

$\begin{vmatrix} -h & -1 \\ -1 & -h \end{vmatrix} = 0$

$h^2 - 1 = 0$

$h^2 = 1$

$h = -1$ or 1

Sağ taraftan U ile çarparsak

$\boxed{H.v = (I - 2.uu^T).U \longrightarrow -u}$

$h = -1$ için

$(H - hI).V = 0$

$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix} = 0$

$\downarrow$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$h = 1$ için

$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$

$\downarrow$

$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

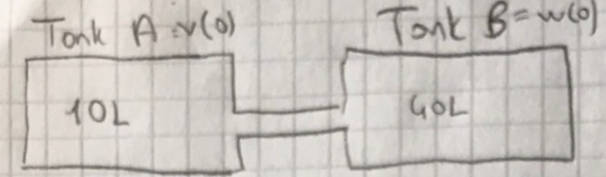
d) Is H diagonalizable? Why or why not?

Yes it is diagonalizable.

$U = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ $\quad H = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

$H = 1 - 2.U.U^T$ ✓

4)



a) $\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$ is correct equation.

1

1 1 ✓

b and c)

$$\begin{vmatrix} 2-\lambda & -2 \\ -1 & 1-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(1-\lambda) - 2 = 0$$

$$2 - 2\lambda - \lambda + \lambda^2 - 2 = 0$$

$$\lambda^2 - 3\lambda = 0$$

$$\boxed{\lambda = 0 \quad \lambda = 3}$$

$\lambda = 0$ için $(A - \lambda I) \cdot v = 0$

$$\begin{vmatrix} 2 & -2 \\ -1 & 1 \end{vmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda = 3$ için $(A - \lambda I) \cdot v = 0$

$$\begin{bmatrix} -1 & -2 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\downarrow$$

$$\begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$u(t) = S_1 \cdot e^{\lambda_1 t} c_1 + S_2 e^{\lambda_2 t}$$

$$u(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cancel{e^{0}} \cdot c_1 + \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^{3t} c_2$$

b) t = 1 sec için

$$u(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} c_1 + \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^{3} c_2$$

c) Steady state

$t \to \infty$

$$u(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} c_1 + \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^{\infty} \cdot c_2$$

$\downarrow$

$e^{-\infty}$ olsaydı her şey çok güzel olacaktı...